Edexcel International A-Level Physics Unit 4

Topic 5A: Momentum, Collisions, and Impulse

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1. Momentum and Newton's Laws in 2D

Momentum ($\vec{p}$) is a fundamental vector quantity in physics, defined as the product of an object's mass (m) and its velocity ($\vec{v}$).

$$p = mv$$

- **SI Unit:** Kilogram meters per second (kg m s^{-1}) or Newton-seconds (N s).
- **Vector Nature:** Momentum possesses both magnitude and direction. It always points in the exact same direction as the velocity vector.

Newton's Second Law Redefined

In classical mechanics, Newton's Second Law is most accurately stated in terms of momentum: *The net external force acting on an object is directly equal to the rate of change of its linear momentum.*

$$F = \frac{\Delta p}{\Delta t}$$

When mass remains constant during motion, this expression smoothly transitions into the familiar $F = ma$:

$$F = \frac{\Delta(mv)}{\Delta t} = m\left(\frac{\Delta v}{\Delta t}\right) = ma$$

The Principle of Conservation of Momentum

The **Law of Conservation of Linear Momentum** states that for a closed system (a system isolated from external forces like friction or gravity), the total momentum before an interaction is identically equal to the total momentum after the interaction.

$$\Sigma\vec{p}_{\text{initial}} = \Sigma\vec{p}_{\text{final}}$$

Handling Momentum in Two Dimensions (2D)

When objects collide at angles or scatter off-course, momentum must be treated as a vector split into two independent, perpendicular axes: the horizontal (x-axis) and the vertical (y-axis).

To solve 2D momentum problems easily, resolve the velocities into components using simple trigonometry:

- **Horizontal component (x):** $v_x = v\cos\theta$
- **Vertical component (y):** $v_y = v\sin\theta$

The conservation law holds completely true for each individual axis:

$$\text{Total Initial } p_x = \text{Total Final } p_x$$

$$\text{Total Initial } p_y = \text{Total Final } p_y$$

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2. Elastic and Inelastic Collisions

While total momentum and total energy are unconditionally conserved in all closed-system interactions, the behavior of **Kinetic Energy (E_k)** determines the category of the collision.

Collision Type	Momentum	Total Energy	Kinetic Energy (E_k)	Physical Characteristics
Elastic	Conserved	Conserved	**Conserved**	No energy is lost as heat, sound, or permanent deformation.
Inelastic	Conserved	Conserved	**Not Conserved**	Some E_k transforms into thermal energy, sound, or work done deforming objects.
Perfectly Inelastic	Conserved	Conserved	**Maximum Loss**	The colliding objects stick together completely and move with a shared final velocity.

Kinetic Energy Formulas for Momentum Problems

When mass and velocity are directly known:

$$E_k = \frac{1}{2}mv^2$$

When working directly with momentum values (p), it is much quicker to calculate kinetic energy using:

$$E_k = \frac{p^2}{2m}$$

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3. Impulse

Impulse (Δp) represents the measure of how a force changes the momentum of an object over a period of time.

$$\text{Impulse} = F\Delta t = \Delta p = m\vec{v} - m\vec{u}$$

- **Units:** N s or kg m s^{-1}.
- **Force-Time Graphs:** Because forces during impacts (like a bat hitting a ball) vary over time, impulse is graphically evaluated as the **area under a Force-time (F-t) graph**.

Engineering Safety Applications

Rearranging the impulse equation highlights how force depends on impact time:

$$F = \frac{\Delta p}{\Delta t}$$

If a moving object has a fixed change in momentum (Δp) to come to a complete stop, **increasing the duration of the impact (Δt) drastically reduces the average force (F) experienced.**

- **Vehicle Crumple Zones:** Deliberately collapse to prolong the crash duration, lowering deceleration forces on passengers.
- **Airbags:** Increase the time over which a passenger's head comes to rest, preventing high-force skull impacts.

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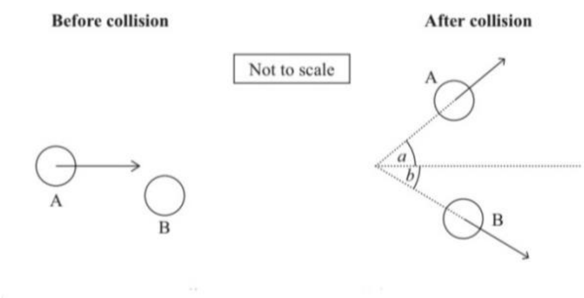
4. Exam-Style Worked Examples

Context Diagram and Data

Two identical pucks, A and B, undergo a 2D collision. Puck B is initially stationary.



- **Mass of each puck (m):** 0.110 kg
- **Initial momentum of A (p_i):** 0.046 kg m s^{-1}
- **Final momentum of A (p_A):** 0.039 kg m s^{-1}
- **Angle of deflection for A (a):** $33°$

Question (i)

> **Deduce whether the angle between the paths of the pucks after the collision was $90°$. You should use the principle of conservation of momentum.**

Step 1: Set up the Vertical (y-axis) Conservation Equation

Initially, Puck A moves purely horizontally and Puck B is stationary. The total initial vertical momentum is exactly zero.

$$\text{Total Initial } p_y = \text{Total Final } p_y$$

$$0 = p_A \sin(a) - p_B \sin(b)$$

$$p_A \sin(a) = p_B \sin(b)$$

Substitute the known values for Puck A:

$$0.039 \times \sin(33°) = p_B \sin(b)$$

$$0.02124 \text{ kg m s}^{-1} = p_B \sin(b) \quad \text{— [Equation 1]}$$

Step 2: Set up the Horizontal (x-axis) Conservation Equation

The total initial horizontal momentum is purely the initial momentum of Puck A.

$$\text{Total Initial } p_x = \text{Total Final } p_x$$

$$0.046 = p_A \cos(a) + p_B \cos(b)$$

Substitute the known values:

$$0.046 = 0.039 \times \cos(33^\circ) + p_B \cos(b)$$

$$0.046 = 0.03271 + p_B \cos(b)$$

$$0.046 - 0.03271 = p_B \cos(b)$$

$$0.01329 \text{ kg m s}^{-1} = p_B \cos(b) \quad \text{— [Equation 2]}$$

Step 3: Calculate Angle b

Divide Equation 1 by Equation 2. This cancels out the unknown final momentum p_B and leaves us with $\tan(b)$:

$$\frac{p_B \sin(b)}{p_B \cos(b)} = \frac{0.02124}{0.01329}$$

$$\tan(b) = 1.5982$$

$$b = \tan^{-1}(1.5982) = 57.97^\circ \approx 58^\circ$$

Step 4: Find the total angle between the final paths

$$\text{Total Angle} = a + b = 33° + 57.97° = 90.97°$$

Conclusion: Since $90.97° \approx 90°$, we deduce that within reasonable experimental uncertainty limits, the angle between the paths of the pucks after the collision is indeed $90°$.

Question (ii)

Deduce whether the collision was elastic.

$$\text{mass of each puck} = 0.110 \text{ kg}$$

To determine if the collision is elastic, we must verify if the total kinetic energy before the collision equals the total kinetic energy after the collision. We will use the formula $E_k = \frac{p^2}{2m}$.

Step 1: Calculate Total Initial Kinetic Energy

$$\text{Initial } E_k = \frac{(p_i)^2}{2m} = \frac{(0.046)^2}{2 \times 0.110} = \frac{0.002116}{0.220} = 0.00962 \text{ J}$$

Step 2: Determine the Final Momentum of Puck B (p_B)

Using Equation 1 from our previous work ($0.02124 = p_B \sin(57.97°)$):

$$p_B = \frac{0.02124}{\sin(57.97°)} = \frac{0.02124}{0.8478} = 0.0251 \text{ kg m s}^{-1}$$

Step 3: Calculate Total Final Kinetic Energy

$$\text{Final } E_k = E_{k,\text{A}} + E_{k,\text{B}}$$

$$\text{Final } E_k = \frac{(p_A)^2}{2m} + \frac{(p_B)^2}{2m}$$

$$\text{Final } E_k = \frac{(0.039)^2}{2 \times 0.110} + \frac{(0.0251)^2}{2 \times 0.110}$$

$$\text{Final } E_k = \frac{0.001521}{0.220} + \frac{0.000630}{0.220}$$

$$\text{Final } E_k = 0.00691\text{ J} + 0.00286\text{ J} = 0.00977\text{ J}$$

Step 4: Compare Initial and Final Kinetic Energy Values

- Initial $E_k = 0.00962$ J
- Final $E_k = 0.00977$ J

Conclusion:
The minor difference between the initial and final values (0.00015 J) is entirely down to the rounding of angles and values throughout the steps. Because Initial $E_k \approx$ Final E_k, kinetic energy is conserved. The collision was **elastic**.

Topic 5B: Circular Motion and Applications

1. Kinematics of Circular Motion

Circular motion describes an object traveling along a curved path at a constant radius r. Even if the object moves at a constant tangential speed, its direction changes continuously. Because velocity ($\vec{v}$) is a vector quantity involving both speed and direction, a continuous change in direction means the object is constantly accelerating.

- **Angular Displacement (θ):** The angle swept out by the radius vector as an object moves along a circular arc, measured in radians (rad).
- **Angular Velocity (ω):** The rate of change of angular displacement with respect to time.

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T} = 2\pi f$$

  Where T is the period (time for one complete revolution) and f is the frequency of rotation.

- **Linear Velocity (v):** The instantaneous velocity directed tangentially to the circular path. The relationship between linear speed and angular speed is given by:

$$v = r\omega$$

2. Centripetal Acceleration and Centripetal Force

Since the velocity vector changes direction toward the center of the path, the object experiences a **centripetal acceleration** (a_c). This acceleration always points directly toward the center of rotation.

$$a_c = \frac{v^2}{r} = r\omega^2$$

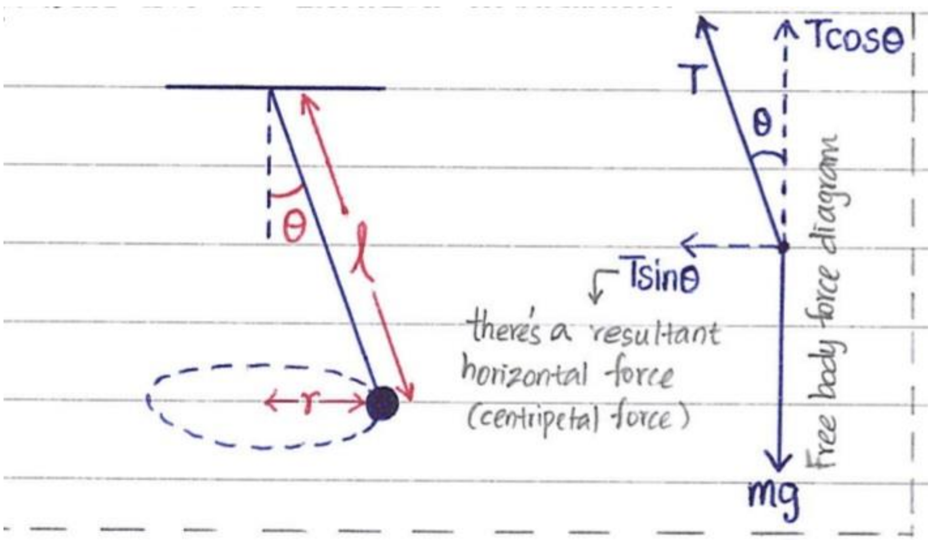
According to Newton's Second Law ($F = ma$), a net external force must act on the object to cause this acceleration. This net inward force is defined as the **centripetal force** (F_c).

$$F_c = \frac{mv^2}{r} = mr\omega^2$$

> **Conceptual Note:** Centripetal force is **not** a separate physical force. It is simply a label given to the *net resultant force* pointing toward the center of the circle, which can be provided by tension, friction, gravity, normal reaction, or a combination of these forces.

3. Application 1: The Conical Pendulum

A conical pendulum consists of a mass m attached to a string of length L, swiveling in a horizontal circle of radius r at a constant angular velocity ω. The string sweeps out the surface of a cone, making an angle θ with the vertical axis.



Force Analysis

Two main forces act on the mass m:

1. **Weight (mg):** Acting vertically downwards.
2. **Tension (T):** Acting along the string at an angle θ to the vertical.

Resolving the tension vector into perpendicular components:

- **Vertical component:** $T\cos\theta$
- **Horizontal component:** $T\sin\theta$

Since there is no acceleration along the vertical axis, the vertical forces must balance perfectly:

$$T\cos\theta = mg \quad \text{— [Equation 1]}$$

The horizontal component of the tension points directly toward the center of the horizontal circular path, providing the necessary centripetal force:

$$T\sin\theta = mr\omega^2 \quad \text{— [Equation 2]}$$

Derivation of Angular Velocity (ω)

From the geometry of the system, the radius r of the circle relates to the string length L by:

$$r = l\sin\theta$$

Substitute this expression for r into Equation 2:

$$T\sin\theta = m(l\sin\theta)\omega^2$$

$$T = ml\omega^2 \quad \text{— [Equation 3]}$$

Now substitute Equation 3 into Equation 1:

$$(ml\omega^2)\cos\theta = mg$$

$$\omega^2\cos\theta = \frac{g}{l}$$

$$\omega = \sqrt{\frac{g}{L\cos\theta}}$$

Why the String Can Never Be Horizontal

For the string to become completely horizontal, the angle θ must reach 90°.

Let us analyze the vertical balance equation: $T\cos\theta = mg$. If $\theta = 90^\circ$, then $\cos(90^\circ) = 0$. The equation becomes:

$$T(0) = mg \Longrightarrow 0 = mg$$

Since the mass m and the gravitational field strength g are non-zero positive values, this statement is a physical impossibility ($0 \neq mg$). Expressed dynamically:

$$T = \frac{mg}{\cos\theta}$$

As $\theta \to 90^\circ$, $\cos\theta \to 0$, which causes the tension T to approach infinity (∞) just to support the weight of the mass. Therefore, a string spinning a mass on Earth can never be perfectly horizontal.

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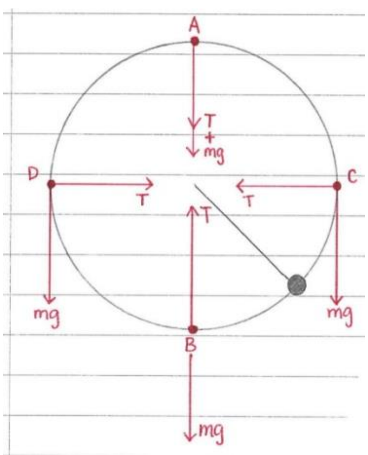
4. Application 2: Mass on a String in a Vertical Circle

When a mass moves in a vertical circle under the influence of gravity, its linear speed v varies continuously, making it non-uniform circular motion. We analyze the tension T at the two critical boundary states: the top and the bottom of the loop.



At the Highest Point (Top of the Circle)

At the top of the circle, both the tension (T_{top}) and the weight (mg) point vertically downwards toward the center of the circular path. Together, they form the net centripetal force:

$$F_c = T_{\text{top}} + mg = \frac{mv_{\text{top}}^2}{r}$$

$$T_{\text{top}} = \frac{mv_{\text{top}}^2}{r} - mg$$

At the Lowest Point (Bottom of the Circle)

At the bottom of the circle, the tension (T_{bottom}) pulls vertically upward toward the center, while the weight (mg) pulls downwards away from the center. The net force directed toward the center is:

$$F_c = T_{\text{bottom}} - mg = \frac{mv_{\text{bottom}}^2}{r}$$

$$T_{\text{bottom}} = \frac{mv_{\text{bottom}}^2}{r} + mg$$

> **Key Takeaway:** The string experiences maximum tension at the lowest point of the motion, making this the position where the string is most susceptible to breaking.

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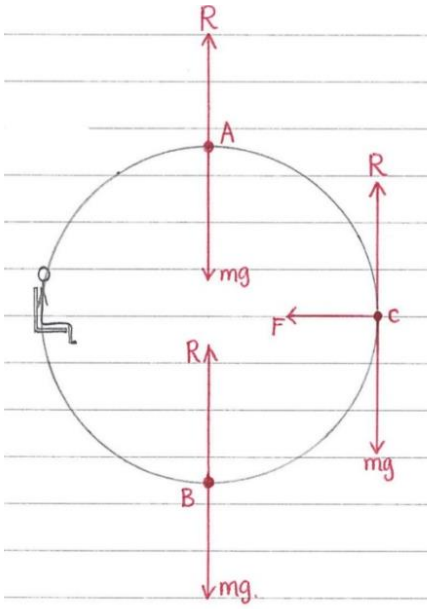
5. Application 3: The Ferris Wheel

A Ferris wheel carries passengers at a constant linear speed v around a vertical circle of radius r. The passenger cars pivot to remain upright throughout the rotation, meaning the normal reaction force (R) from the seat always acts vertically upward.



At the Highest Point (Top of the Wheel)

The weight (mg) acts downward toward the center of the wheel, and the normal reaction force (R_{top}) acts upward away from the center. The net force acting toward the center provides the centripetal force:

$$F_c = mg - R_{\text{top}} = \frac{mv^2}{r}$$

$$R_{\text{top}} = mg - \frac{mv^2}{r}$$

As the speed v increases, the normal reaction force decreases, causing the passenger to feel lighter at the peak.

At the Lowest Point (Bottom of the Wheel)

The normal reaction force (R_{bottom}) acts upward toward the center, while the weight (mg) acts downward away from the center. The net force toward the center is:

$$F_c = R_{\text{bottom}} - mg = \frac{mv^2}{r}$$

$$R_{\text{bottom}} = mg + \frac{mv^2}{r}$$

At the bottom, $R_{\text{bottom}} > mg$, meaning the passenger experiences an increased apparent weight, feeling pressed firmly into the seat.

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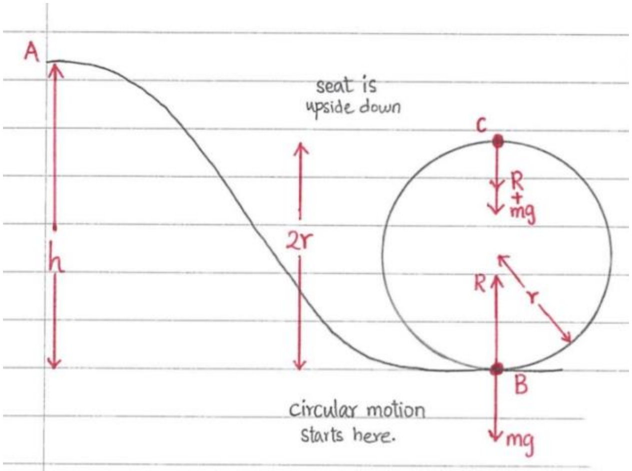
6. Application 4: The Roller Coaster Loop-the-Loop

Unlike a Ferris wheel, a roller coaster cart moves along a fixed track and changes its orientation, turning completely upside down at the top of a vertical loop of radius r.



At the Bottom of the Loop

The track is underneath the cart. The normal reaction force (R_{bottom}) pushes upward toward the center, while the weight (mg) pulls downward away from the center:

$$F_c = R_{\text{bottom}} - mg = \frac{mv^2}{r}$$

$$R_{\text{bottom}} = \frac{mv^2}{r} + mg$$

At the Top of the Loop

The cart is inverted at the apex, meaning the track is located directly above the cart. Both the weight (mg) and the normal reaction force (R_{top}) push downward toward the center of the loop:

$$F_c = R_{\text{top}} + mg = \frac{mv^2}{r}$$

$$R_{\text{top}} = \frac{mv^2}{r} - mg$$

Critical Velocity (v_c)

The **critical velocity** (v_c) is the absolute minimum linear speed required for the roller coaster cart to successfully traverse the top of the vertical loop without losing physical contact with the track.

If the cart travels too slowly, it will lose contact and fall. The boundary condition for just maintaining contact occurs when the normal reaction force between the wheels and the track drops precisely to zero ($R_{\text{top}} = 0$).

Setting $R_{\text{top}} = 0$ in the top equation:

$$0 = \frac{mv_c^2}{r} - mg$$

$$mg = \frac{mv_c^2}{r}$$

Canceling the mass m from both sides of the expression:

$$g = \frac{v_c^2}{r}$$

$$v_c^2 = rg$$

$$v_c = \sqrt{rg}$$